

Week 3: Running the Programs in VS Code

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Guide to the week 3 programs

About this guide. It explains how to run the week 3 programs in **Visual Studio Code**, and gives a short description of each file in the folder `code/week03/slides`. Each program file also starts with a comment that says what it teaches and what output to expect: run it, then compare your result with that comment.

1 What you need

The same setup as the previous weeks: **Flutter** (it includes the Dart SDK, so the `dart` command works), and **Visual Studio Code** with the **Flutter** extension. If something is missing, follow the install guide `001_install_vscode` and the week 1 guide *Running the programs in VS Code*. All week 3 programs are plain Dart programs: no Flutter project is needed.

2 Get the week 3 code and open it

If you already copied the repository, update it: open a terminal in the repository folder and type

```
git pull
```

Otherwise download it with `git clone https://github.com/maoude/mobile_pro_2026.git`. In VS Code choose **File** → **Open Folder...** and open `code/week03`. The files are in the folder `slides`.

3 Run a program

With the **Run** button

Open the file from the Explorer, then click **Run** in the small **Run | Debug** line above `void main()` (or press `F5`; `Ctrl+F5` runs without the debugger). The output appears in the **Debug Console** at the bottom.

With the terminal

Open a terminal with **Terminal** → **New Terminal** (`Ctrl+``), go to the folder and use `dart run`:

```
cd code/week03/slides
dart run 01_simple_class.dart
```

This week two programs use a second file. `02_private_fields.dart` and `03_tostring_and_named_constructor.dart` start with the line `import 'circle.dart'`. Because of it, `circle.dart` must stay in the same folder. Run the programs that start with a number; `circle.dart` has no main and is not run by itself.

4 The programs

Each entry gives the file, what it shows, and the output you should get.

1. A simple class

`01_simple_class.dart`

A `Circle` class with a field, a constructor and a method. The constructor shorthand `Circle(this.radius)` stores the argument in the field. An object is created without `new`, and every field has an implicit getter and setter.

```
Radius: 10.0
Area: 314.1592653589793
New radius: 20.0
New area: 1256.6370614359173
```

2. The Circle library

`circle.dart`

The complete `Circle` class, in its own file. It has a private field `_radius` (private means private to the *file*), a getter and a setter that refuses a radius that is not positive, a `toString` method, and a named constructor `Circle.create()`. It is imported by the next two programs, and it prints nothing by itself.

3. Private fields, getters and setters

`02_private_fields.dart`

Uses `Circle` from another file. It reads and changes the radius through the getter and the setter, catches the exception thrown for a negative radius, and shows that the constructor is protected too.

```
Radius: 10.0
After c1.radius = 20: 20.0
Exception: Invalid Radius
Radius is still: 20.0
Constructor also validates: Exception: Invalid Radius
```

4. toString and named constructors

`03_tostring_and_named_constructor.dart`

Printing an object calls its `toString`. A second circle is created with the named constructor, which always has radius 5.

```
c1 -> Radius: 10.0 Area: 314.1592653589793
c2 -> Radius: 5.0 Area: 78.53981633974483
Same text with toString(): Radius: 5.0 Area: 78.53981633974483
```

5. Inheritance

`04_inheritance.dart`

`Circle` extends `Shape`: the subclass constructor calls `super(color)`, and `toString` reuses the parent's text with `super.toString()`. The program also tests a type with `is` and prints a list that holds both a `Shape` and a `Circle`.

```
Shape: Color: green
Circle: Color: red Radius: 10.0
This shape is a plain Shape
This shape is a Circle with radius 10.0
All shapes in one list:
Color: green
Color: red Radius: 10.0
```

5 Quick troubleshooting

Problem	What to do
<code>dart</code> is not recognised	Close all terminals and restart VS Code so it sees the new PATH; check with <code>flutter doctor</code> .
Error about <code>circle.dart</code> not found	The file must be in the same folder as the program that imports it. Open the folder <code>code/week03/slides</code> , or run the program from that folder.
Running <code>circle.dart</code> says there is no main	That is normal: it is a library. Run <code>02_private_fields.dart</code> instead.
The week 3 folder is missing	You have an old copy: run <code>git pull</code> in the repository folder.

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